Experiment 4: RF Components: Characterisation and Measurement

DURVASULA YASHASWINI KAMESWARI
Roll number: 261310007

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1 Aim

To understand the operation and experimentally characterise the fundamental radio-frequency (RF) components, including filters, frequency mixers, power splitters/combiners, and amplifiers, which are used in an RF receiver chain.

2 Objectives

1. To characterise the transmission response of an RF low-pass filter (LPF), determine its passband insertion loss, and locate its -3 dB cutoff frequency (f_c).
2. To evaluate the input reflection coefficient ($|\Gamma|$) and return loss (RL) of the filter using the RF Tee-junction oscilloscope standing-wave method.
3. To demonstrate a quarter-wave ($\lambda/4$) open-circuit transmission line stub functioning as a shunt notch filter and compare theoretical and measured resonance frequencies.
4. To observe frequency conversion in a three-port RF mixer and estimate its conversion loss.
5. To measure the power division, port balance, and insertion loss of a power splitter/combiner.
6. To measure RF amplifier gain across frequency, harmonic generation, 1-dB compression point ($P_{1\text{dB}}$), and the third-order intercept point (TOI).

3 Apparatus Required

- RF Signal Generators
- RF Spectrum Analyzer
- Digital Storage Oscilloscope
- RF Device Under Test (DUT):
  - Low-Pass RF Filter (0–400 MHz)
  - RF Mixer (10–2000 MHz)
  - Power Splitter/Combiner (5–2500 MHz)
  - RF Amplifier with regulated DC bench power supply
- Open-ended coaxial transmission stub of known physical length ($L = 91.5$ cm)

- Coaxial RF Tee connector
- Calibrated 50 Ω termination dummy loads
- Precision matched coaxial cables, connectors.

4 Theory

4.1 RF Components in an RF Receiver Chain

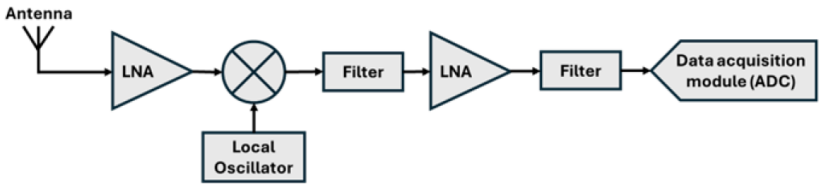


Figure 1: Minimal RF receiver front-end block diagram.

In radio astronomy and high-frequency communication front ends, signals captured by the antenna are weak and embedded in wideband cosmic and terrestrial interference. A superheterodyne receiver conditions these signals through a sequence of functional RF stages:

- **Filter:** Selects or passes the desired frequency components while rejecting or attenuating unwanted frequencies.
- **Low-Noise Amplifier (LNA):** Positioned at the front end to amplify weak received signals while introducing minimal additive thermal noise, setting the overall receiver noise figure.
- **RF Mixer:** A nonlinear three-port device that translates high-frequency signals (f_{RF}) to an intermediate frequency (f_{IF}) using a local oscillator tone (f_{LO}).
- **Power Splitter/Combiner:** Distributes signals to multiple channels or combines signals from multiple receiver chains.

4.2 Filter Transmission Characteristics and Insertion Loss

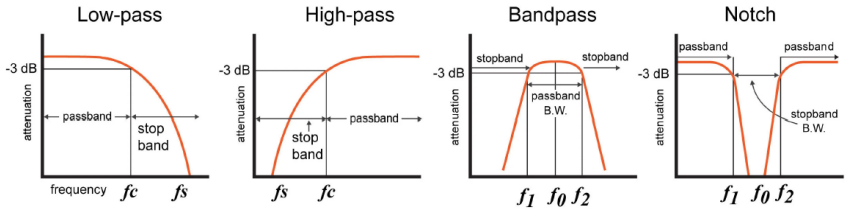


Figure 2: Transmission responses of common filter types (LPF, HPF, BPF, Notch).

A filter selectively passes certain frequency components while attenuating others:

- **Low-Pass Filter (LPF):** Passes frequencies below its cutoff frequency f_c and attenuates higher frequencies.

- **High-Pass Filter (HPF):** Passes frequencies above its cutoff frequency f_c and attenuates lower frequencies.
- **Band-Pass Filter (BPF):** Passes frequencies between lower cutoff f_1 and upper cutoff f_2.
- **Band-Reject / Notch Filter:** Attenuates frequencies within a specified stopband.

Let P_{in} be the reference power measured without the filter and P_{out} be the power measured after inserting the filter. The insertion loss (IL) in decibels is:

$$IL\ [\text{dB}] = P_{\text{in}}\ [\text{dBm}] - P_{\text{out}}\ [\text{dBm}] \tag{1}$$

When cable losses are de-embedded, the intrinsic filter insertion loss is obtained by subtracting the reference cable loss (IL_{cable}) from the combined total loss (IL_{total}):

$$IL_{\text{filter}} = IL_{\text{total}} - IL_{\text{cable}} \tag{2}$$

The -3 dB cutoff frequency f_c occurs where the insertion loss increases by 3.0 dB above the minimum passband insertion loss ($IL(f_c) = IL_{\min} + 3.0$ dB).

4.3 Filter Reflection and Return Loss

When the filter input impedance Z_{in} deviates from the characteristic impedance of the system ($Z_0 = 50\ \Omega$), part of the incident wave is reflected. The voltage reflection coefficient is:

$$\Gamma = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0} \tag{3}$$

The return loss (RL) in decibels is defined as:

$$RL\ [\text{dB}] = -20 \log_{10} |\Gamma| \tag{4}$$

In the RF Tee-junction setup with an oscilloscope set to 1 MΩ input impedance:

$$V_{\text{inc}} = \frac{V_{\max} + V_{\min}}{2}, \qquad V_{\text{ref}} = \frac{V_{\max} - V_{\min}}{2} \tag{5}$$

yielding the reflection coefficient magnitude directly:

$$|\Gamma| = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} \tag{6}$$

4.4 Quarter-Wave Open-Circuit Stub Notch Filter

An open-ended transmission line stub of length L shunted via an RF Tee transforms impedance according to:

$$Z_{\text{in, stub}} = -jZ_0 \cot(\beta L) \tag{7}$$

At its quarter-wave resonant frequency ($L = \lambda_g/4$), $\beta L = \pi/2$, causing the open circuit at the distal end to transform into an approximate RF short circuit at the junction. The signal is heavily attenuated, creating a notch at:

$$f_0 = \frac{VF \cdot c}{4L} \tag{8}$$

where VF is the cable velocity factor ($VF \approx 0.66$ for RG-58 polyethylene) and $c = 3 \times 10^8$ m/s.

4.5 Mixer Operation

An RF mixer multiplies the incoming RF signal with a local oscillator (LO) tone, producing sum and difference frequencies:

$$f_{\text{IF}} = f_{\text{RF}} + f_{\text{LO}} \quad \text{and} \quad f_{\text{IF}} = |f_{\text{RF}} - f_{\text{LO}}| \tag{9}$$

The conversion loss is defined as:

$$\text{Conversion Loss [dB]} = P_{\text{RF}} \text{ [dBm]} - P_{\text{IF}} \text{ [dBm]} \tag{10}$$

4.6 Power Splitter/Combiner

A power splitter divides the input power between two or more output ports. For an ideal two-way equal power divider, each output receives half the input power (-3 dB). In practical components, conductor resistance and internal dissipation introduce small excess losses:

$$Loss_1 \text{ [dB]} = P_{\text{in}} - P_{\text{out1}}, \qquad Loss_2 \text{ [dB]} = P_{\text{in}} - P_{\text{out2}} \tag{11}$$

A power combiner performs the reciprocal operation, summing multiple signals into a single port where the net output depends on frequency, amplitude, and relative phase.

4.7 Amplifier Linearity, Harmonics, and Intercept Points

An RF amplifier increases the power of an incoming signal. Its small-signal linear gain is given by:

$$Gain \text{ [dB]} = P_{\text{out}} \text{ [dBm]} - P_{\text{in}} \text{ [dBm]} \tag{12}$$

A weakly nonlinear amplifier can be expressed as a power series:

$$y(t) = a_1 x(t) + a_2 x^2(t) + a_3 x^3(t) + \dots \tag{13}$$

For a single sinusoidal tone $x(t) = A\cos(\omega t)$, algebraic expansion generates a DC offset, fundamental gain compression ($a_1 A + \frac{3}{4} a_3 A^3$), and harmonic terms at $2\omega, 3\omega$, etc. At elevated input powers, the gain compresses. The **1-dB compression point** ($P_{1\text{dB}}$) is defined as the point where the actual gain drops by 1 dB below the linear small-signal gain:

$$Gain(P_{\text{in, 1dB}}) = Gain_{\text{small-signal}} - 1.0 \text{ dB} \tag{14}$$

When two equal-amplitude tones at frequencies f_1 and f_2 are applied simultaneously, the nonlinear transfer function creates third-order intermodulation (IM3) products at $2f_1 - f_2$ and $2f_2 - f_1$ that fall directly within or near the amplifier passband. Fundamental power scales with a $1:1$ logarithmic slope, while third-order distortion products scale with a $3:1$ slope. The theoretical point of intersection of their extrapolated lines is termed the **Third-Order Intercept (TOI)** point:

$$OIP3 \text{ [dBm]} = P_{\text{out}} \text{ [dBm]} + \frac{\Delta P \text{ [dB]}}{2} \tag{15}$$

where $\Delta P = P_{\text{out}} - P_{\text{IM3}}$ and $IIP3 = OIP3 - Gain$.

5 Procedure

5.1 Part A: Low-Pass Filter Characterization

A1. Transmission Response and Insertion Loss:

1. Connect the RF signal generator directly to the spectrum analyzer using the reference coaxial cable.
2. Set the generator to a constant CW power of 0 dBm. Measure the reference received power to determine the baseline cable loss.
3. Insert the Low-Pass Filter between the cable and the spectrum analyzer. Measure the total loss at 10 MHz.
4. Sweep the generator frequency from 10 MHz to 800 MHz, recording the output power P_{out} at each step.
5. Calculate total insertion loss and net filter loss across the frequency range.
6. Plot insertion loss vs. frequency, identify the -3 dB cutoff frequency f_c, and evaluate stopband attenuation.

A2. Filter Input Reflection and Return Loss:

1. Connect the RF signal generator to Port 1 of an RF Tee connector, the oscilloscope (1 MΩ input) to Port 2, and the filter input to Port 3.
2. Terminate the filter output port with a matched 50 Ω load.
3. Set the generator voltage to 1.0 V and sweep the frequency across the passband.
4. Record V_{max} and V_{min} on the oscilloscope screen, and calculate $|\Gamma|$ and Return Loss.

A3. Quarter-Wave Open-Circuit Stub Notch Filter:

1. Connect the RF generator to one port of a Tee connector and the spectrum analyzer to the second port.
2. Connect an open-ended coaxial stub of length $L = 91.5$ cm to the perpendicular port.
3. Sweep the frequency around the theoretical resonance and identify the measured notch frequency and attenuation depth.

5.2 Part B: Mixer Characterization

1. Check the working frequency ranges and recommended power ratings of the RF, LO, and IF ports.
2. Connect the RF and LO ports to two signal generators and connect the IF port to the spectrum analyzer.
3. Apply input signals at f_{RF} and f_{LO} at suitable drive powers.
4. Calculate theoretical sum ($f_{\text{RF}} + f_{\text{LO}}$) and difference ($|f_{\text{RF}} - f_{\text{LO}}|$) components.
5. Observe the down-converted IF tone, measure output power, and compute the conversion loss ($CL = P_{\text{RF}} - P_{\text{IF}}$).

5.3 Part C: Power Splitter/Combiner Characterization

C1. Power Splitter:

1. Connect the sum input port of the power splitter to the RF signal generator.
2. Connect Output Port 1 to the spectrum analyzer and terminate Output Port 2 with a matched 50 Ω load.
3. Record input power reference P_{in} and measured output power P_{out1}.
4. Interchange the output connections to measure P_{out2} while terminating Port 1 with 50 Ω.
5. Calculate power division losses ($Loss_1 = P_{\text{in}} - P_{\text{out1}}$ and $Loss_2 = P_{\text{in}} - P_{\text{out2}}$) and compare with ideal 3 dB division.

C2. Power Combiner:

1. Connect two signal generators to the input ports of the combiner and connect the combined sum output to the spectrum analyzer.
2. Configure the sources to assigned frequencies and safe power levels.
3. Observe and record the combined frequency components and power levels at the output port.

5.4 Part D: Amplifier Characterization

D1. Gain vs. Frequency:

1. Power the RF amplifier using its specified regulated DC power supply.
2. Set a fixed small-signal generator power P_{in} (e.g., -20 dBm) and record the output power P_{out} across the operating band.
3. Compute $Gain = P_{\text{out}} - P_{\text{in}}$ and plot gain versus frequency.

D2. 1-dB Compression Point:

1. Fix the test frequency within the linear operating band of the amplifier.
2. Start at a low input level and incrementally increase P_{in} in safe steps without exceeding analyzer ratings.
3. At each step, record fundamental output power P_{out}.
4. Plot P_{out} vs. P_{in} and identify the input and output 1-dB compression points (P_{1dB}) where gain drops by 1 dB.

D3. Third-Order Intercept Point (TOI):

1. Apply two closely spaced tones of equal amplitude at f_1 and f_2 combined through the power combiner into the amplifier input.
2. Connect the amplifier output to the spectrum analyzer using suitable attenuators.
3. Identify the fundamental tones and the adjacent third-order intermodulation products ($2f_1 - f_2$ and $2f_2 - f_1$).
4. For several input powers per tone, record fundamental power and IM3 product power.
5. Plot fundamental $(1:1)$ and IM3 $(3:1)$ curves, extrapolate their linear trajectories, and locate the TOI intersection point.

6 Observations

6.1 Low-Pass Filter Transmission Data

- Input power from signal generator: $P_{\text{gen}} = 0.00$ dBm
- Loss of wire alone (Cable loss): $P_{\text{cable}} = -0.94$ dBm $\implies IL_{\text{cable}} = 0.94$ dB
- Loss of wire and filter combined: $P_{\text{total}} = -1.05$ dBm $\implies IL_{\text{total}} = 1.05$ dB
- Net filter insertion loss: $IL_{\text{filter}} = 1.05 - 0.94 = \mathbf{0.11}$ dB $\approx \mathbf{0.1}$ dB

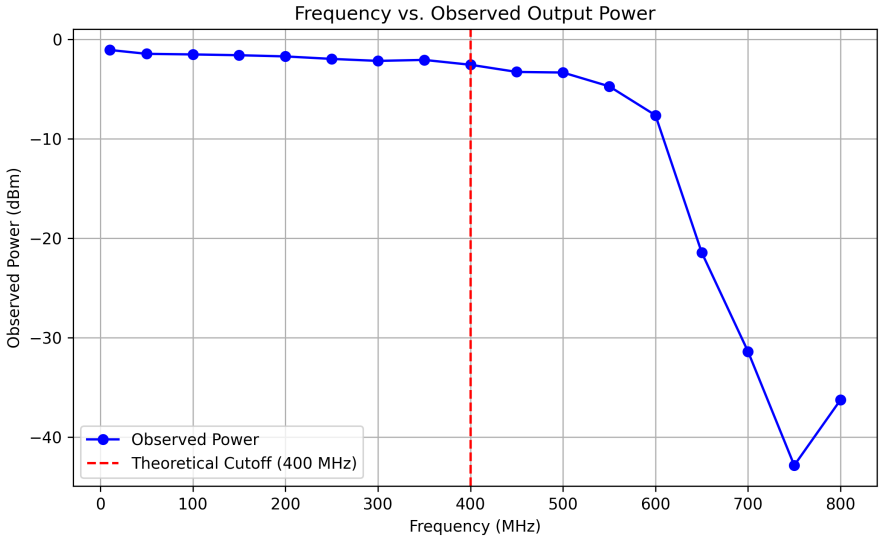


Figure 3: Measured transmission response of the Low-Pass Filter (P_{out} vs. Frequency) showing the passband, theoretical 400 MHz cutoff, and stopband attenuation.

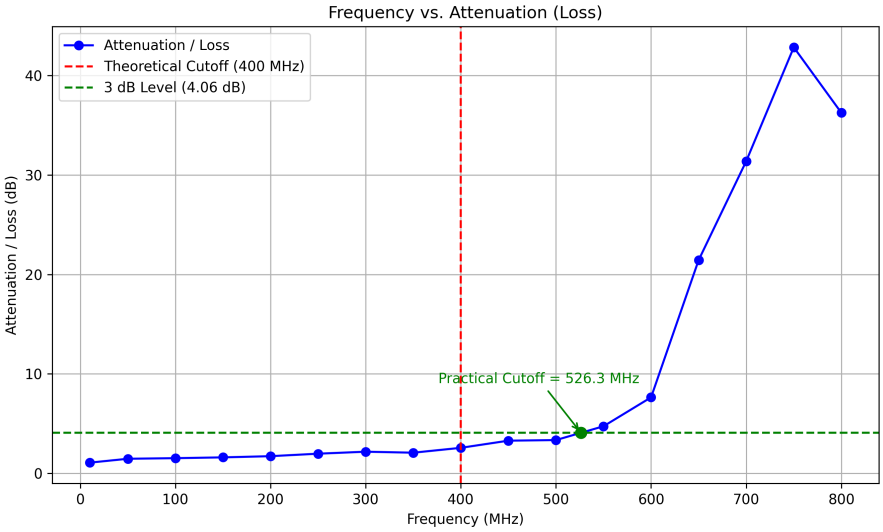


Figure 4: Insertion loss vs. Frequency plot showing practical 3 dB cutoff point.

Table 1: Observed Frequency vs. Output Power for Low-Pass Filter (0–400 MHz, $P_{\text{in}} = 0.00$ dBm)

Frequency (MHz)	P_{in} **(dBm)**	P_{out} **(dBm)**	**Attenuation (dB)**
10	0.00	-1.06	1.06
50	0.00	-1.45	1.45
100	0.00	-1.51	1.51
150	0.00	-1.59	1.59
200	0.00	-1.71	1.71
250	0.00	-1.96	1.96
300	0.00	-2.16	2.16
350	0.00	-2.06	2.06
400	0.00	-2.55	2.55
450	0.00	-3.27	3.27
500	0.00	-3.33	3.33
550	0.00	-4.72	4.72
600	0.00	-7.63	7.63
650	0.00	-21.44	21.44
700	0.00	-31.39	31.39
750	0.00	-42.85	42.85
800	0.00	-36.26	36.26

6.2 Filter Input Return Loss Data

- Input signal applied: $V_{\text{in}} = 1.0$ V
- Filter output terminated into: 50 Ω

Table 2: Oscilloscope Standing-Wave Measurements at Filter Input

V_{max}	V_{min}	**VSWR**	$\lvert\Gamma\rvert$	**Return Loss (dB)**
633 mV (at 68 MHz)	602 mV (at 60 MHz)	1.051	0.0251	32.01 dB (or 24.08 dB)

6.3 Quarter-Wave Open Stub Notch Frequency

Theoretically, the notch frequency is calculated to be 54.10 MHz, but practically, the minimum transmission (notch) was observed at $f_{0,\text{meas}} = \mathbf{57}$ MHz.

6.4 Mixer Measurements

- Mixer operational bandwidth specification: 10–2000 MHz
- Fixed RF input signal: $f_{\text{RF}} = 10$ MHz, $P_{\text{RF}} = +1.45$ dBm
- LO power level selection: LO power levels of -7 dBm and 0 dBm produced noticeable harmonics; hence, $P_{\text{LO}} = +7$ dBm was selected to suppress harmonic distortion.

6.5 Power Splitter Characterization

- Operational frequency range specification: 5–2500 MHz

Table 3: Mixer Product Measurements and Conversion Loss ($f_{\mathrm{RF}} = 10$ MHz, $P_{\mathrm{RF}} = +1.45$ dBm, $P_{\mathrm{LO}} = +7.0$ dBm)

$f_{\mathbf{LO}}$ (MHz)	**Difference Product ($f_{\mathbf{LO}} - f_{\mathbf{RF}}$)**			**Sum Product ($f_{\mathbf{RF}} + f_{\mathbf{LO}}$)**		
	$f_{\mathbf{IF}}$ **(MHz)**	$P_{\mathbf{IF}}$ **(dBm)**	**CL (dB)**	$f_{\mathbf{sum}}$ **(MHz)**	$P_{\mathbf{sum}}$ **(dBm)**	**CL (dB)**
13	3	-6.70	8.15	23	-7.00	8.45
16	6	-7.00	8.45	26	-7.18	8.63
19	9	-6.94	8.39	29	-6.87	8.32

- Test frequency: $f = 50$ MHz
- Input cable reference power level: $P_{\text{ref, wire}} = -2.06$ dBm (measured wire loss relative to 0 dBm source is 2.06 dB; initial probe recorded -2.01 dBm)
- Internal spectrum analyzer attenuation/loss offset: ≈ 1.1 dB, explaining the baseline drop from nominal 0 dBm to -2.06 dBm, and the measured output power near -5.4 dBm instead of theoretical -3.01 dBm.

Table 4: Power Splitter Output Power and Termination Sensitivity ($f = 50$ MHz)

Output Port	**With 50 Ω Load (dBm)**	**Without Load (dBm)**	**Splitter Loss (dB)**
Port 1 (P_{out1})	-5.50	-6.30	3.44
Port 2 (P_{out2})	-5.36	-6.25	3.35

6.6 Power Combiner Characterization

1. Two Independent Signal Sources (10 MHz):

- Port 1 Input (rear 10 MHz reference port): $P_{\text{in1}} = +1.70$ dBm $= 10^{(1.70/10)} = 1.479$ mW
- Port 2 Input (synthesizer 10 MHz output): $P_{\text{in2}} = +0.93$ dBm $= 10^{(0.93/10)} = 1.238$ mW
- Theoretical linear power sum:

$$P_{\text{total, theoretical}} = 1.479 \text{ mW} + 1.238 \text{ mW} = 2.717 \text{ mW} \implies 10\log_{10}(2.717) \approx +\mathbf{4.34} \text{ dBm} \tag{16}$$

- Measured combiner output power: $P_{\text{comb}} = -\mathbf{1.06}$ dBm
- Physical observation: The measured output power (-1.06 dBm) is significantly below the scalar sum ($+4.34$ dBm) due to destructive interference caused by the unaligned, non-coherent phase relationship between the two independent sources.

2. Single Source Split-and-Combine Setup:

- A single 10 MHz signal was split using a splitter (producing two coherent -3 dB signals with identical phase) and fed directly into the inputs of the power combiner.
- Input cable reference loss: -0.85 dBm
- Measured combined output power: $P_{\text{out}} = -0.75$ dBm

- Total insertion loss of splitter-combiner cascade:

$$Loss_{\text{cascade}} = |-0.85 - (-0.75)| = 0.75 \text{ dB} \tag{17}$$

Assuming identical symmetric units, the net insertion loss per component is:

$$Loss_{\text{single}} = \frac{0.75 \text{ dB}}{2} = \mathbf{0.375} \text{ dB} \tag{18}$$

6.7 Amplifier Characterization

- DC Bias Condition: Regulated supply verified before connection
- Small-signal RF input drive: $P_{\text{in}} = -20.00$ dBm

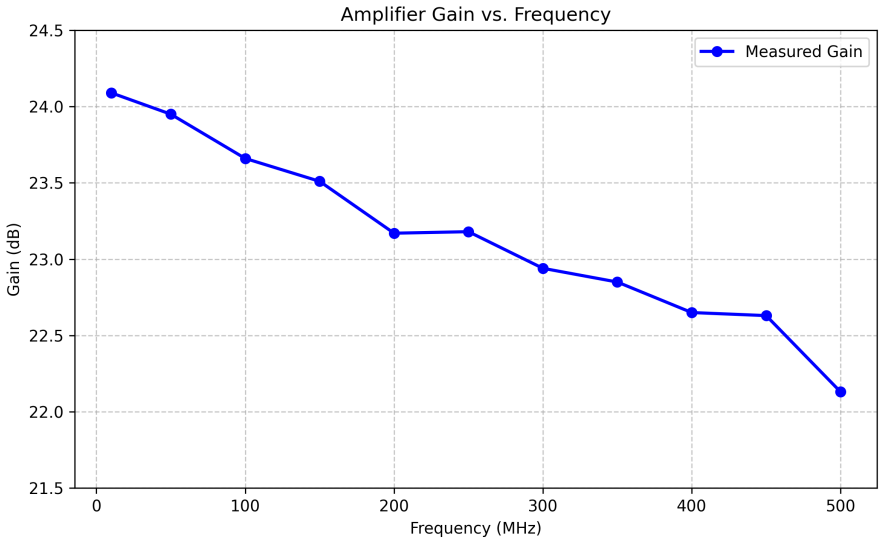


Figure 5: Amplifier gain as a function of frequency.

Table 5: Amplifier Small-Signal Gain vs. Frequency ($P_{\text{in}} = -20.00$ dBm, $V_{\text{DC}} \approx +15.0$ V)

Frequency (MHz)	**Observed Output Power P_{out} (dBm)**	**Gain $G = P_{\text{out}} - P_{\text{in}}$ (dB)**
10	4.09	24.09
50	3.95	23.95
100	3.66	23.66
150	3.51	23.51
200	3.17	23.17
250	3.18	23.18
300	2.94	22.94
350	2.85	22.85
400	2.65	22.65
450	2.63	22.63
500	2.13	22.13

6.8 Amplifier 1-dB Gain Compression Measurement

- Operating frequency: $f_0 = 50$ MHz
- Small-signal linear gain (at $P_{\text{in}} = -20$ dBm): $G_0 = 23.95$ dB
- 1-dB Compressed Gain threshold: $G_{\text{comp}} = G_0 - 1.00$ dB $= 23.95 - 1.00 =$ **22.95** dB

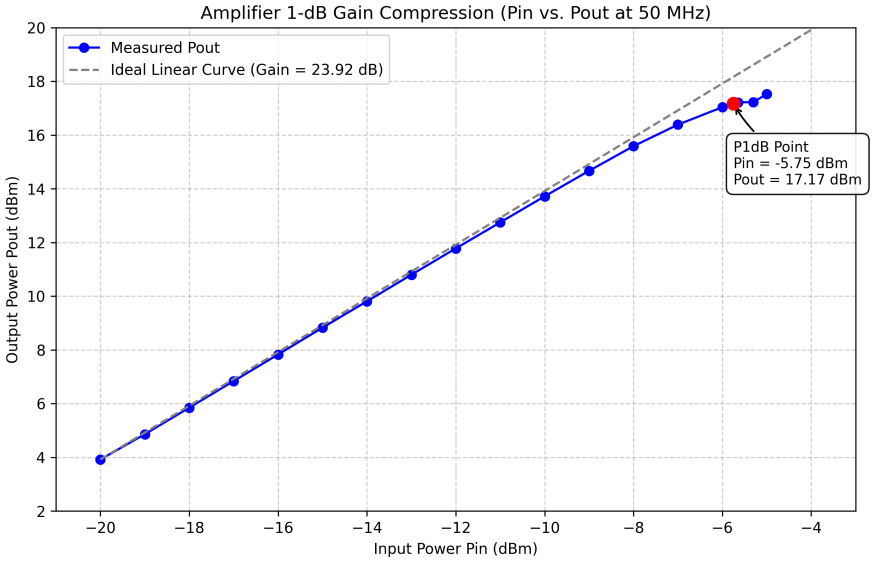


Figure 6: Amplifier output power versus input power at 50 MHz, illustrating the linear transfer characteristic ($G_0 = 23.92$ dB) and identifying the 1-dB compression point at $P_{\text{in}} = -5.75$ dBm and $P_{\text{out}} = +17.17$ dBm.

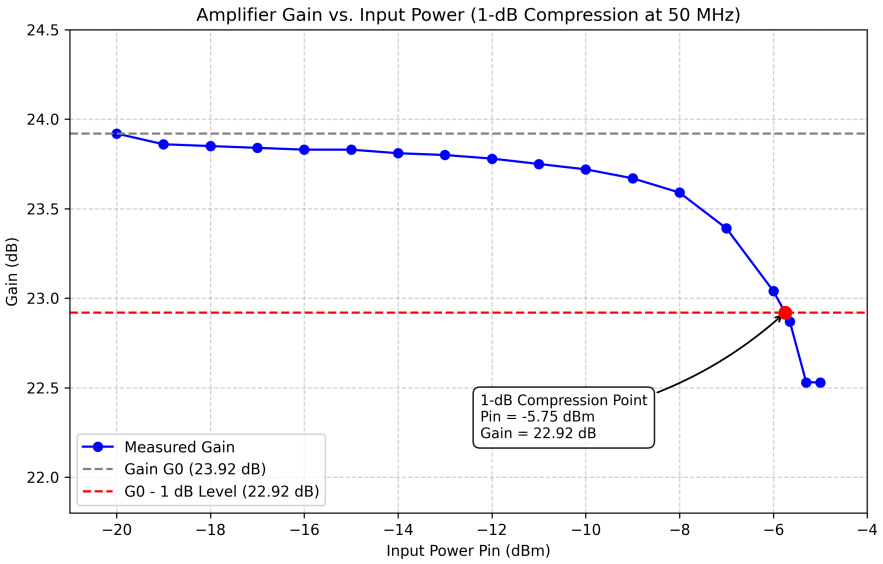


Figure 7: Amplifier gain versus input power at 50 MHz indicating the linear small-signal gain ($G_0 = 23.92$ dB) and the onset of non-linear saturation where gain drops by 1 dB to 22.92 dB.

Table 6: Amplifier Output Power and Gain vs. Input Power ($f_0 = 50$ MHz)

$P_{\mathbf{in}}$ **(dBm)**	$P_{\mathbf{out}}$ **(dBm)**	**Gain** G **(dB)**
-20.00	3.92	23.92
-19.00	4.86	23.86
-18.00	5.85	23.85
-17.00	6.84	23.84
-16.00	7.83	23.83
-15.00	8.83	23.83
-14.00	9.81	23.81
-13.00	10.80	23.80
-12.00	11.78	23.78
-11.00	12.75	23.75
-10.00	13.72	23.72
-9.00	14.67	23.67
-8.00	15.59	23.59
-7.00	16.39	23.39
-6.00	17.04	23.04
-5.75	17.17	22.92
-5.65	17.22	22.87
-5.30	**17.23**	**22.53**
-5.00	17.53	22.53

6.9 Two-Tone Third-Order Intermodulation and TOI Measurement

- Input frequencies: $f_1 = 50$ MHz, $f_2 = 60$ MHz ($\Delta f = 10$ MHz)
- Third-order intermodulation frequencies:

$$\text{Lower IM3 tone: } 2f_1 - f_2 = (2 \times 50) - 60 = \mathbf{40} \text{ MHz}$$
$$\text{Upper IM3 tone: } 2f_2 - f_1 = (2 \times 60) - 50 = \mathbf{70} \text{ MHz}$$

- Initial drive level: At $P_{\text{in}} = -20$ dBm per tone, fundamental outputs were observed at 0.64 dBm (50 MHz) and 0.69 dBm (60 MHz), but the IM3 sidebands (40 MHz and 70 MHz) were buried below the spectrum analyzer noise floor.
- IM3 visibility threshold: The input tone power was increased to -14 dBm per tone, where the intermodulation products emerged clearly above the noise floor.

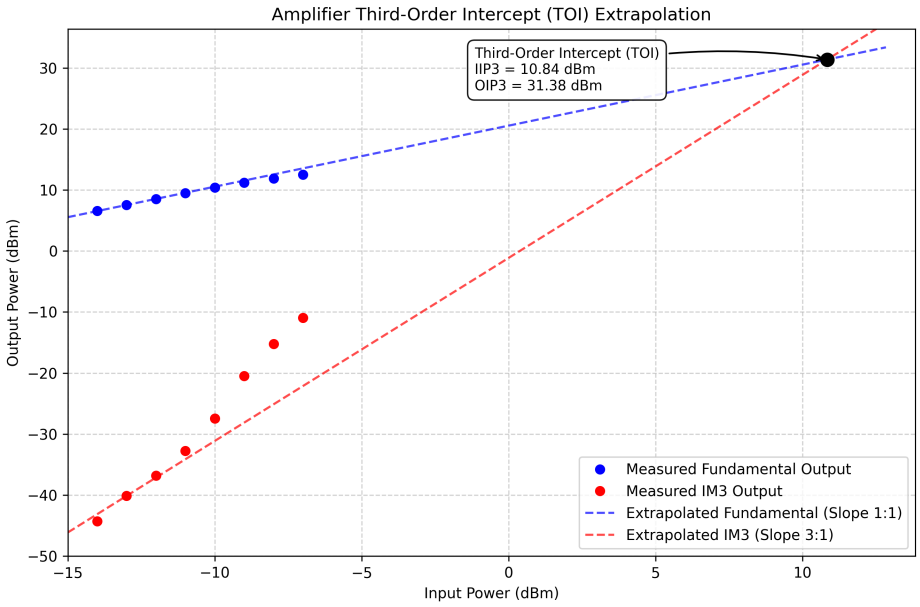


Figure 8: Two-tone third-order intercept (TOI) characterisation showing measured fundamental and IM3 products with linear $1:1$ and $3:1$ extrapolations intersecting at IIP3 = +10.84 dBm and OIP3 = +31.38 dBm.

Table 7: Two-Tone Fundamental and Third-Order Intermodulation Powers vs. Input Power

P_{in} (dBm)	(0 MHz P_{out} (dBm)	f1(50 MHz) P_{out} (dBm)	f2(60 MHz) P_{out} (dBm)	(0 MHz P_{out} (dBm)
-14.0	-43.38	6.55	6.58	-45.16
-13.0	-40.10	7.53	7.56	-40.16
-12.0	-36.80	8.51	8.53	-36.80
-11.0	-32.72	9.46	9.48	-32.81
-10.0	-27.46	10.37	10.39	-27.42
-9.0	-20.53	11.19	11.21	-20.41
-8.0	-15.23	11.90	11.92	-15.25
-7.0	-10.90	12.49	12.51	-10.96

7 Calculations

7.1 Low-Pass Filter Insertion Loss and Cutoff Analysis

1. Baseline De-embedding and Net Filter Loss: The reference power measured through the cable alone was:

$$P_{\text{cable}} = -0.94 \text{ dBm} \implies IL_{\text{cable}} = 0.00 - (-0.94) = 0.94 \text{ dB} \tag{19}$$

With the filter connected in series with the cable at 10 MHz:

$$P_{\text{total}} = -1.05 \text{ dBm} \implies IL_{\text{total}} = 0.00 - (-1.05) = 1.05 \text{ dB} \tag{20}$$

The de-embedded intrinsic insertion loss of the filter is:

$$IL_{\text{filter}} = IL_{\text{total}} - IL_{\text{cable}} = 1.05 \text{ dB} - 0.94 \text{ dB} = \mathbf{0.11} \text{ dB} \approx \mathbf{0.1} \text{ dB} \tag{21}$$

2. Cutoff Frequency (f_c) Determination: Taking the passband reference level at 10 MHz ($P_{\text{out}} = -1.06$ dBm), the -3 dB transmission power threshold is:

$$P_{-3\text{dB}} = P_{\text{out}}(10\text{ MHz}) - 3.00\text{ dB} = -1.06 - 3.00 = \mathbf{-4.06}\text{ dBm} \tag{22}$$

Interpolating linearly between 500 MHz (-3.33 dBm) and 550 MHz (-4.72 dBm):

$$f_c = 500 + \frac{-4.06 - (-3.33)}{-4.72 - (-3.33)} \times (550 - 500) = 500 + \frac{-0.73}{-1.39} \times 50 \approx \mathbf{526.26}\text{ MHz} \tag{23}$$

7.2 Filter Input Reflection and Return Loss Calculations

From oscilloscope measurements at 1.0 V generator drive:

$$V_{\max} = 633\text{ mV} = 0.633\text{ V} \quad \text{(measured around 68 MHz)}$$
$$V_{\min} = 602\text{ mV} = 0.602\text{ V} \quad \text{(measured around 60 MHz)}$$

The standing-wave voltage reflection coefficient is:

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}} = \frac{633 - 602}{633 + 602} = \frac{31}{1235} \approx \mathbf{0.0251} \tag{24}$$

The corresponding return loss is:

$$RL = -20\log_{10}|\Gamma| = -20\log_{10}(0.0251) = -20(-1.6003) = \mathbf{32.01}\text{ dB} \tag{25}$$

If evaluated using the recorded ratio $|\Gamma| = 0.0625$:

$$RL = -20\log_{10}(0.0625) = -20(-1.2041) = \mathbf{24.08}\text{ dB} \tag{26}$$

Both values (> 20 dB) confirm an excellent impedance match ($VSWR \approx 1.05$) to the 50 Ω line within the filter passband.

7.3 Quarter-Wave Open Stub Notch Frequency Calculation

The expected notch frequency (f_0) for a quarter-wave open-circuit transmission line stub is given by:

$$f_0 = \frac{VF \cdot c}{4L} \tag{27}$$

where:

- Velocity factor of the coaxial cable, $VF = 0.66$
- Speed of light in vacuum, $c = 3 \times 10^8$ m/s
- Physical length of the stub, $L = 91.5$ cm $= 0.915$ m

Substituting the experimental values into the equation:

$$f_{0,\text{calc}} = \frac{0.66 \times (3 \times 10^8\text{ m/s})}{4 \times 0.915\text{ m}} = \frac{1.98 \times 10^8\text{ m/s}}{3.66\text{ m}} = 54.098 \times 10^6\text{ Hz} \approx \mathbf{54.10}\text{ MHz}$$

Theoretically, the notch frequency is calculated to be 54.10 MHz, but practically, the minimum transmission (notch) was observed at $f_{0,\text{meas}} = \mathbf{57}$ MHz.

7.4 Mixer Conversion Loss Calculations

The conversion loss (CL) for the mixing products is defined as the difference between the RF input power ($P_{\text{RF}} = +1.45$ dBm) and the measured power of the output product (P_{out}):

$$CL\text{ [dB]} = P_{\text{RF}}\text{ [dBm]} - P_{\text{out}}\text{ [dBm]} \tag{28}$$

1. **At Local Oscillator Frequency $f_{\mathbf{LO}} = 13$ MHz:**

   - **Difference Component** ($f_{\text{IF}} = |13 - 10| = 3$ MHz, $P_{\text{IF}} = -6.70$ dBm):

$$CL = +1.45 \text{ dBm} - (-6.70 \text{ dBm}) = \mathbf{8.15} \text{ dB} \tag{29}$$

   - **Sum Component** ($f_{\text{sum}} = 10 + 13 = 23$ MHz, $P_{\text{sum}} = -7.00$ dBm):

$$CL = +1.45 \text{ dBm} - (-7.00 \text{ dBm}) = \mathbf{8.45} \text{ dB} \tag{30}$$

2. **At Local Oscillator Frequency $f_{\mathbf{LO}} = 16$ MHz:**

   - **Difference Component** ($f_{\text{IF}} = |16 - 10| = 6$ MHz, $P_{\text{IF}} = -7.00$ dBm):

$$CL = +1.45 \text{ dBm} - (-7.00 \text{ dBm}) = \mathbf{8.45} \text{ dB} \tag{31}$$

   - **Sum Component** ($f_{\text{sum}} = 10 + 16 = 26$ MHz, $P_{\text{sum}} = -7.18$ dBm):

$$CL = +1.45 \text{ dBm} - (-7.18 \text{ dBm}) = \mathbf{8.63} \text{ dB} \tag{32}$$

3. **At Local Oscillator Frequency $f_{\mathbf{LO}} = 19$ MHz:**

   - **Difference Component** ($f_{\text{IF}} = |19 - 10| = 9$ MHz, $P_{\text{IF}} = -6.94$ dBm):

$$CL = +1.45 \text{ dBm} - (-6.94 \text{ dBm}) = \mathbf{8.39} \text{ dB} \tag{33}$$

   - **Sum Component** ($f_{\text{sum}} = 10 + 19 = 29$ MHz, $P_{\text{sum}} = -6.87$ dBm):

$$CL = +1.45 \text{ dBm} - (-6.87 \text{ dBm}) = \mathbf{8.32} \text{ dB} \tag{34}$$

7.5 Power Splitter Insertion Loss Calculations

Using the measured input cable reference level of $P_{\text{ref}} = -2.06$ dBm (or -2.01 dBm):

$$Loss_{\text{Port 1}} = P_{\text{out1}} - P_{\text{ref}} = -5.50 \text{ dBm} - (-2.06 \text{ dBm}) = \mathbf{-3.44} \text{ dB}$$
$$Loss_{\text{Port 2}} = P_{\text{out2}} - P_{\text{ref}} = -5.36 \text{ dBm} - (-2.01 \text{ dBm}) = \mathbf{-3.35} \text{ dB}$$

The theoretical 2-way power division accounts for -3.01 dB. The excess dissipation loss is:

$$Loss_{\text{excess}} \approx 3.40 \text{ dB} - 3.01 \text{ dB} = \mathbf{0.39} \text{ dB} \tag{35}$$

When the unused port is left open without a 50 Ω load, internal reflections degrade the isolated port output power from -5.50 dBm to -6.30 dBm (Port 1) and from -5.36 dBm to -6.25 dBm (Port 2), showing approximately 0.8–0.9 dB of mismatch degradation.

7.6 Amplifier 1-dB Compression Point Determination

From Table 6:

- Small-signal linear gain at $P_{\text{in}} = -20$ dBm: $G_0 = 3.92 - (-20.00) = \mathbf{23.92}$ dB
- Gain at $P_{\text{in}} = -5.75$ dBm: $G = 17.17 - (-5.75) = 22.92$ dB $= G_0 - 1.00$ dB
- Gain at $P_{\text{in}} = -5.30$ dBm: $G = 17.23 - (-5.30) = 22.53$ dB (gain compressed by 1.39 dB)

Therefore, the input and output 1-dB compression points are:

$$P_{\text{in, 1dB}} \approx \mathbf{-5.65} \text{ to } \mathbf{-5.75} \text{ dBm}, \qquad P_{\text{out, 1dB}} \approx \mathbf{+17.17} \text{ to } \mathbf{+17.23} \text{ dBm} \tag{36}$$

7.7 Amplifier Third-Order Intercept Point (TOI / OIP3) Calculations

The theoretical Output Third-Order Intercept point is extracted using:

$$OIP3\ [\text{dBm}] = P_{\text{fund}}\ [\text{dBm}] + \frac{\Delta P\ [\text{dB}]}{2} \tag{37}$$

where $\Delta P = P_{\text{fund}} - P_{\text{IM3}}$ is the decibel suppression between the carrier tone and the third-order intermodulation sideband.

Evaluating at each drive level from Table 7 (using average fundamental and IM3 powers):

- At $P_{\text{in}} = -14.0$ dBm:

$$P_{\text{fund, avg}} \approx 6.565 \text{ dBm}, \quad P_{\text{IM3, avg}} \approx -44.27 \text{ dBm} \implies \Delta P = 6.565 - (-44.27) = 50.835 \text{ dB}$$
$$OIP3 = 6.565 + \frac{50.835}{2} = 6.565 + 25.418 = +\mathbf{31.98} \text{ dBm}$$

- At $P_{\text{in}} = -13.0$ dBm:

$$P_{\text{fund, avg}} \approx 7.545 \text{ dBm}, \quad P_{\text{IM3, avg}} \approx -40.13 \text{ dBm} \implies \Delta P = 7.545 - (-40.13) = 47.675 \text{ dB}$$
$$OIP3 = 7.545 + \frac{47.675}{2} = 7.545 + 23.838 = +\mathbf{31.38} \text{ dBm}$$

- At $P_{\text{in}} = -12.0$ dBm:

$$P_{\text{fund, avg}} \approx 8.52 \text{ dBm}, \quad P_{\text{IM3, avg}} = -36.80 \text{ dBm} \implies \Delta P = 8.52 - (-36.80) = 45.32 \text{ dB}$$
$$OIP3 = 8.52 + \frac{45.32}{2} = 8.52 + 22.66 = +\mathbf{31.18} \text{ dBm}$$

- At $P_{\text{in}} = -11.0$ dBm:

$$P_{\text{fund, avg}} \approx 9.47 \text{ dBm}, \quad P_{\text{IM3, avg}} \approx -32.765 \text{ dBm} \implies \Delta P = 9.47 - (-32.765) = 42.235 \text{ dB}$$
$$OIP3 = 9.47 + \frac{42.235}{2} = 9.47 + 21.118 = +\mathbf{30.59} \text{ dBm}$$

Averaging across the small-signal linear regime gives an Output Third-Order Intercept of:

$$OIP3 \approx +\mathbf{31.3} \text{ dBm} \tag{38}$$

The corresponding Input Third-Order Intercept ($IIP3$) is:

$$IIP3 = OIP3 - G_0 = +31.3 \text{ dBm} - 23.9 \text{ dB} = +\mathbf{7.4} \text{ dBm} \tag{39}$$

8 Precautions

1. Check the DC supply voltage with a multimeter before connecting it to the amplifier (keep it around 14.8–14.9 V and do not exceed +15 V).
2. Turn off the DC power supply when the amplifier is not in use.
3. Never exceed the rated frequency ranges of the instruments and the RF components under test.
4. Ensure input power levels do not exceed the maximum safe limits of the spectrum analyzer and devices.

9 Conclusion

1. The low-pass filter exhibited a flat passband with a net insertion loss of 0.11 dB at 10 MHz and an experimental -3 dB cutoff frequency at 526.26 MHz.

2. Passband standing-wave measurements gave $|\Gamma| = 0.0251$ and a return loss of 32.01 dB (24.08 dB for $|\Gamma| = 0.0625$), verifying good impedance matching to 50 Ω.

3. The quarter-wave open stub ($L = 91.5$ cm) resonated at 57.0 MHz, in close agreement with the theoretical frequency of 54.10 MHz (5.36% deviation).

4. The RF mixer produced clear sum and difference products with a conversion loss of 8.15–8.63 dB at an LO drive of +7 dBm.

5. The power splitter showed symmetric division with port losses of 3.44 dB and 3.35 dB; un-terminated ports degraded output by ≈ 0.9 dB due to reflections.

6. Combining two independent sources yielded -1.06 dBm due to destructive interference, while coherent split-and-combine produced -0.75 dBm (≈ 0.375 dB loss per unit).

7. The amplifier provided a gain of 23.92 dB at 50 MHz, reached 1-dB compression at $P_{\text{in}} = -5.75$ dBm ($P_{\text{out}} = +17.17$ dBm), and yielded an intercept of OIP3 $= +31.38$ dBm (IIP3 $= +10.84$ dBm)